Home / Grade 8 / Geometry / Quarter 1

Trapezium and the isosceles trapezium

Exactly one pair of parallel sides — and what happens when the two legs are made equal.

LESSON 10–11 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 7–8 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 13' | 8' | 12' | 2'

Warm-up	5 min
Explanation	13 min
Interactive	8 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Name the parts of a trapezium: bases, legs, height.
- Use the fact that the angles on each leg are supplementary.
- Prove and use the properties of an isosceles trapezium.
- Recognise a right-angled trapezium and solve problems using its height.

TEXTBOOK

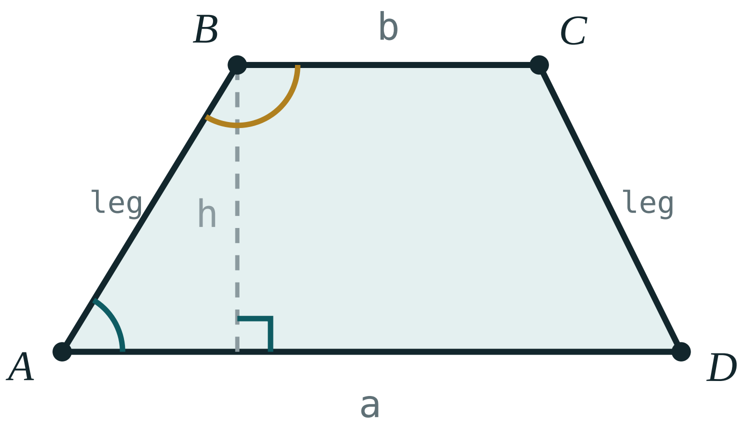
National	Темы 7–8, pp. 19–22
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Definition and parts

DEFINITION

A **trapezium** is a quadrilateral with exactly one pair of parallel sides.



The parallel sides a and b are the bases; the other two are the legs; h is the distance between the bases.

- The parallel sides are the **bases** — the longer one a, the shorter b.
- The other two sides are the **legs**.
- The **height** h is the distance between the two bases, measured perpendicular to them.

ANGLES ON A LEG

The two angles at the ends of a leg are co-interior angles between the parallel bases, so

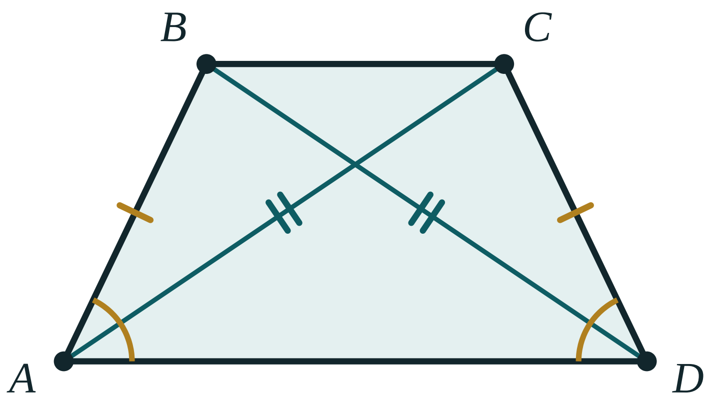
$$\angle A + \angle B = 180^\circ \quad \text{and} \quad \angle C + \angle D = 180^\circ$$

A trapezium with a right angle has two of them (a **right-angled trapezium**), because the angle next to it on the same leg must also be 90°.

The isosceles trapezium

DEFINITION

A trapezium whose two legs are equal is **isosceles**.



AB = CD · ∠A = ∠D · AC = BD

Equal legs, equal base angles and equal diagonals — the three properties travel together.

THEOREM

In an isosceles trapezium:

1. the angles at each base are equal: $\angle A = \angle D$ and $\angle B = \angle C$;
2. the diagonals are equal: $AC = BD$.

Proof of (1). Drop perpendiculars from B and C onto AD. The two right triangles formed have equal hypotenuses (the legs) and equal heights, so they are congruent by RHS — giving $\angle A = \angle D$. The other pair follows from the supplementary angles on each leg.

Proof of (2). $\triangle ABD$ and $\triangle DCA$ have AD common, $AB = DC$, and the included angles $\angle A = \angle D$ — congruent by SAS, so $BD = AC$. ∎

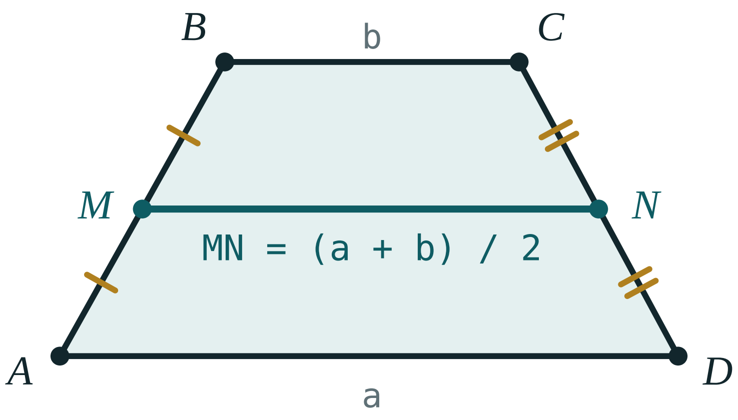
EQUAL DIAGONALS ARE NOT ENOUGH ON THEIR OWN

A rectangle also has equal diagonals. The statement “a trapezium with equal diagonals is isosceles” is true, but the word *trapezium* is doing real work.

The midline (a first look)

The segment joining the midpoints of the two legs is the **midline**. It is parallel to the bases and its length is their average:

$$MN = \frac{a + b}{2}$$



The midline joins the midpoints of the legs and measures exactly the average of the two bases.

The full proof comes in the next lesson but two; for now, use the formula and check it against the interactive model.

Working with the height

Almost every numerical trapezium problem is solved by dropping one or two perpendiculars from the shorter base to the longer one. That cuts the trapezium into a rectangle and one or two right triangles, and then Pythagoras does the rest.

For an **isosceles** trapezium the two triangles are congruent, so each cuts off $\frac{a-b}{2}$ from the longer base — a formula worth remembering.

$$S = \frac{a+b}{2} \cdot h = MN \cdot h$$

02 Worked examples

EXAMPLE 1 ***In trapezium ABCD with $AD \parallel BC$, $\angle A = 65°$. Find $\angle B$.***

1. AB is a leg joining the two parallel bases.

2. $\angle A + \angle B = 180°$

   Co-interior angles.

3. $\angle B = 115°$

ANSWER

$\angle B = 115°$

EXAMPLE 2 **An isosceles trapezium has bases 10 cm and 4 cm and legs 5 cm. Find its height and area.**

1. $\frac{a - b}{2} = \frac{10 - 4}{2} = 3\ cm$

   Each right triangle cuts off 3 cm.

2. $h = \sqrt{5^2 - 3^2} = \sqrt{16} = 4\ cm$

   Pythagoras in one triangle.

3. $MN = \frac{10 + 4}{2} = 7\ cm$

   The midline.

4. $S = 7 \cdot 4 = 28\ cm^2$

ANSWER

$h = 4\ cm,\ S = 28\ cm^2$

EXAMPLE 3 **The midline of a trapezium is 9 cm and one base is 5 cm. Find the other base.**

1. $$\frac{a + b}{2} = 9$$

   The midline formula.

2. $a + b = 18$

3. $a = 18 - 5 = 13\ cm$

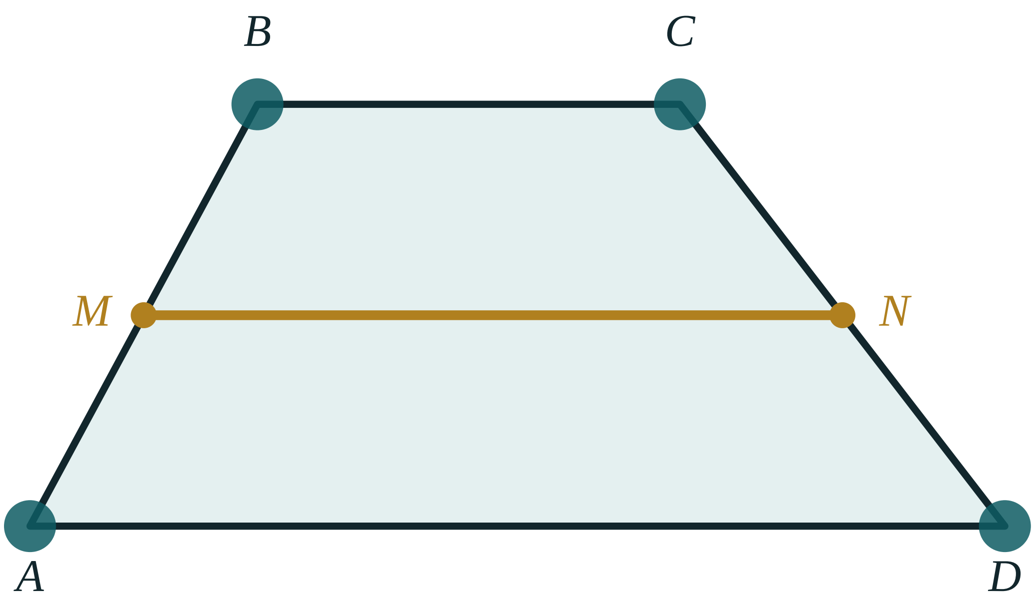
ANSWER

$13\ cm$

03 Interactive model

Drag the corners and read the three numbers: the midline always sits exactly between the two bases.

The midline of a trapezium



a (AD) **300**	b (BC) **130**	(a + b) / 2 **215**	MN measured **215**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Trapezium	Trapetsiya	Трапеция
Base of a trapezium	Trapetsiya asosi	Основание трапеции
Leg (lateral side)	Yon tomon	Боковая сторона
Height	Balandlik	Высота
Isosceles trapezium	Teng yonli trapetsiya	Равнобедренная трапеция
Right-angled trapezium	To‘g‘ri burchakli trapetsiya	Прямоугольная трапеция
Midline	O‘rta chiziq	Средняя линия
Half-sum	Yarim yig‘indi	Полусумма

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A trapezium has:

- both pairs of sides parallel
- exactly one pair of parallel sides
- four equal sides
- perpendicular diagonals

In trapezium ABCD with AD ∥ BC, $\angle A = 72°$. Then $\angle B$ is:

- $72°$
- $108°$
- $18°$
- not determined

The diagonals of an isosceles trapezium are:

- perpendicular
- equal
- bisecting each other
- angle bisectors

The midline of a trapezium with bases 6 and 10 is:

- *4*
- *8*
- *16*
- *60*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *In trapezium ABCD with* $AD \parallel BC$, $\angle A = 65°$. *Find* $\angle B$.

   ANSWER $115°$

2. $\angle C = 105°$. *Find* $\angle D$.

   ANSWER $75°$

3. *The bases are 8 cm and 12 cm. Find the midline.*

   ANSWER *10 cm*

4. *The bases are 5 cm and 9 cm and the height is 4 cm. Find the area.*

   ANSWER *28* cm^2

5. *An isosceles trapezium has* $\angle A = 70°$. *Find* $\angle D$.

   ANSWER $70°$

6. *An isosceles trapezium has* $AC = 13$ *cm. Find BD.*

   ANSWER *13 cm*

7. *A right-angled trapezium has one angle of 90°. How many right angles does it have?*

   ANSWER *2*

Medium · the standard question

7 PROBLEMS

1. *An isosceles trapezium has bases 10 cm and 4 cm and legs 5 cm. Find the height.*

   ANSWER *4 cm*

2. *The same trapezium: find its area.*

   ANSWER *28 cm²*

3. *The midline is 9 cm and one base is 5 cm. Find the other base.*

   ANSWER *13 cm*

4. *An isosceles trapezium has* $\angle A = 60°$*, bases 12 cm and 6 cm. Find the leg.*

   ANSWER $\frac{12 - 6}{2} = 3$, so the leg is $6\ cm$

5. *The area is 48 cm² and the height is 6 cm. Find the midline.*

   ANSWER *8 cm*

6. *An isosceles trapezium has perimeter 34 cm, bases 6 cm and 14 cm. Find each leg.*

   ANSWER *7 cm*

7. *In a right-angled trapezium the bases are 9 cm and 5 cm and the slant leg is 5 cm. Find the height.*

   ANSWER $h = \sqrt{5^2 - 4^2} = 3\ cm$

Hard · stretch and reason

7 PROBLEMS

1. *Prove that a trapezium with equal base angles is isosceles.*

ANSWER Drop the two heights: the right triangles are congruent by ASA, so the legs are equal.

2. *An isosceles trapezium has diagonals that are perpendicular, bases 6 cm and 10 cm. Find its height and area.*

ANSWER Perpendicular diagonals give $h = \frac{a + b}{2} = 8\ cm$ and $S = 8 \cdot 8 = 64\ cm^2$.

3. *In trapezium ABCD, AD ∥ BC, BC = 6 cm, and the diagonal AC bisects ∠A. Prove that AB = BC and find AB.*

ANSWER $\angle CAD = \angle ACB$ (alternate angles) and $\angle CAD = \angle CAB$ (bisector), so $\angle CAB = \angle ACB$ and $\triangle ABC$ is isosceles: $AB = BC = 6\ cm$.

4. *An isosceles trapezium has bases 9 cm and 21 cm and legs 10 cm. Find its height and area.*

ANSWER $\frac{21 - 9}{2} = 6$, so $h = 8\ cm$ and $S = 120\ cm^2$.

5. *Prove that the diagonals of an isosceles trapezium are equal.*

ANSWER $\triangle ABD \cong \triangle DCA$ by SAS: AD common, $AB = DC$, $\angle A = \angle D$.

6. *An isosceles trapezium has perimeter 48 cm, bases 10 cm and 18 cm. Find each leg and the height.*

ANSWER legs $10\ cm$; $\frac{18 - 10}{2} = 4$, so $h = \sqrt{100 - 16} = 2\sqrt{21} \approx 9.2\ cm$.

7. *Show that the height of an isosceles trapezium with bases a, b and leg c is* $\sqrt{c^2 - (\frac{a - b}{2})^2}$

ANSWER Each right triangle has hypotenuse c and horizontal leg $\frac{a - b}{2}$; Pythagoras gives the height.

07 Homework

Homework — 6 problems

Geometry 8, Темы 7–8, pp. 19–22. Draw the height on every figure.

1. *In trapezium ABCD with AD ∥ BC, ∠B = 128°. Find ∠A.*
2. *The bases are 7 cm and 13 cm. Find the midline.*
3. *An isosceles trapezium has bases 16 cm and 6 cm and legs 13 cm. Find its height and area.*
4. *The midline is 11 cm and one base is 8 cm. Find the other.*
5. *An isosceles trapezium has perimeter 44 cm and bases 8 cm and 16 cm. Find each leg.*
6. *Prove that the base angles of an isosceles trapezium are equal.*